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Analiza opóźnień lotów w USA

Opis danych



Źródło: U.S. Department of Transportation (DOT), Bureau of Transportation Statistics



Zawartość:

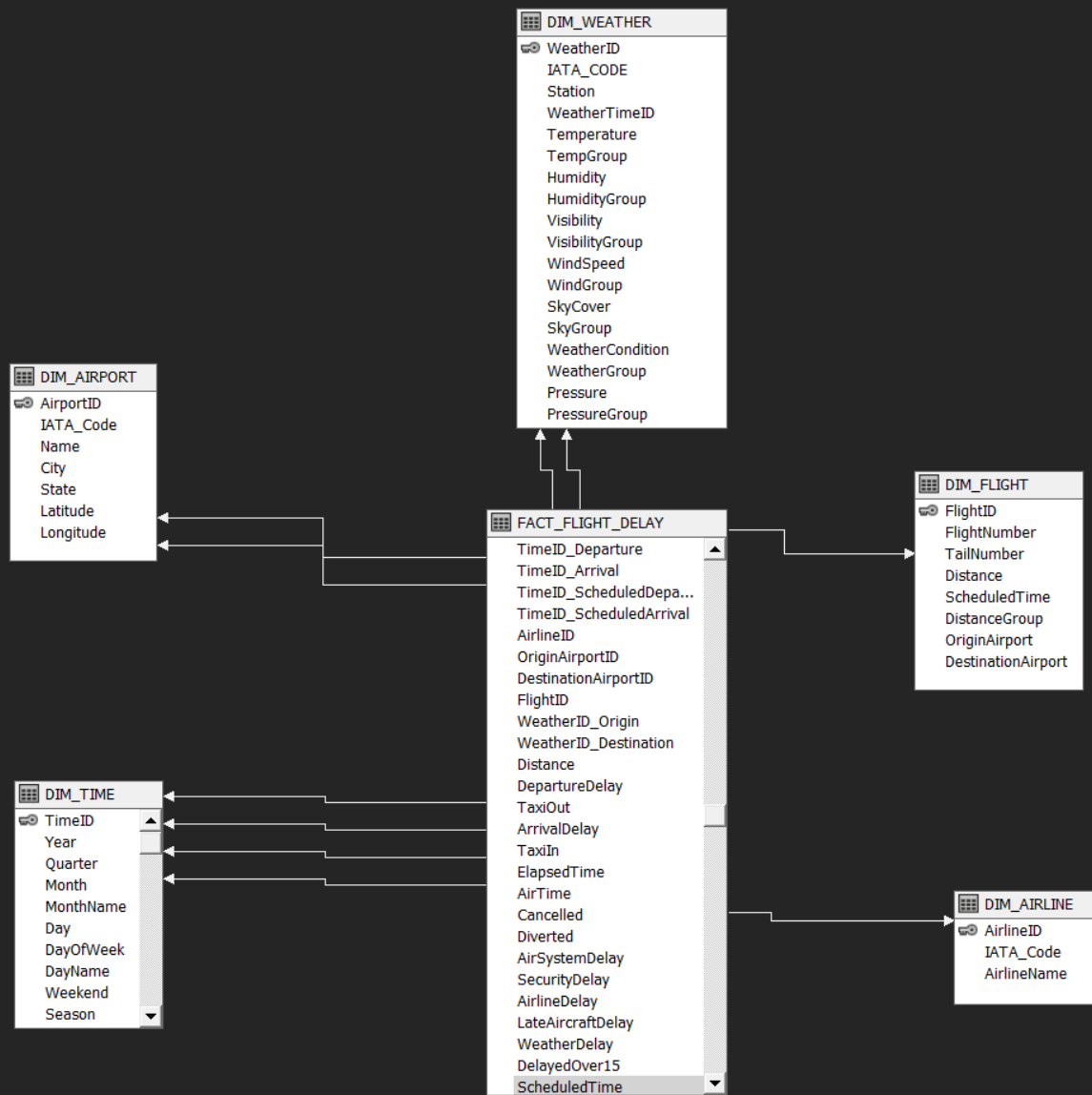
Plik flights.csv – zawiera dane o wszystkich komercyjnych lotach w USA w 2015 r.

Plik airports.csv – dane o lotniskach

Plik airlines.csv – kody i nazwy linii lotniczych

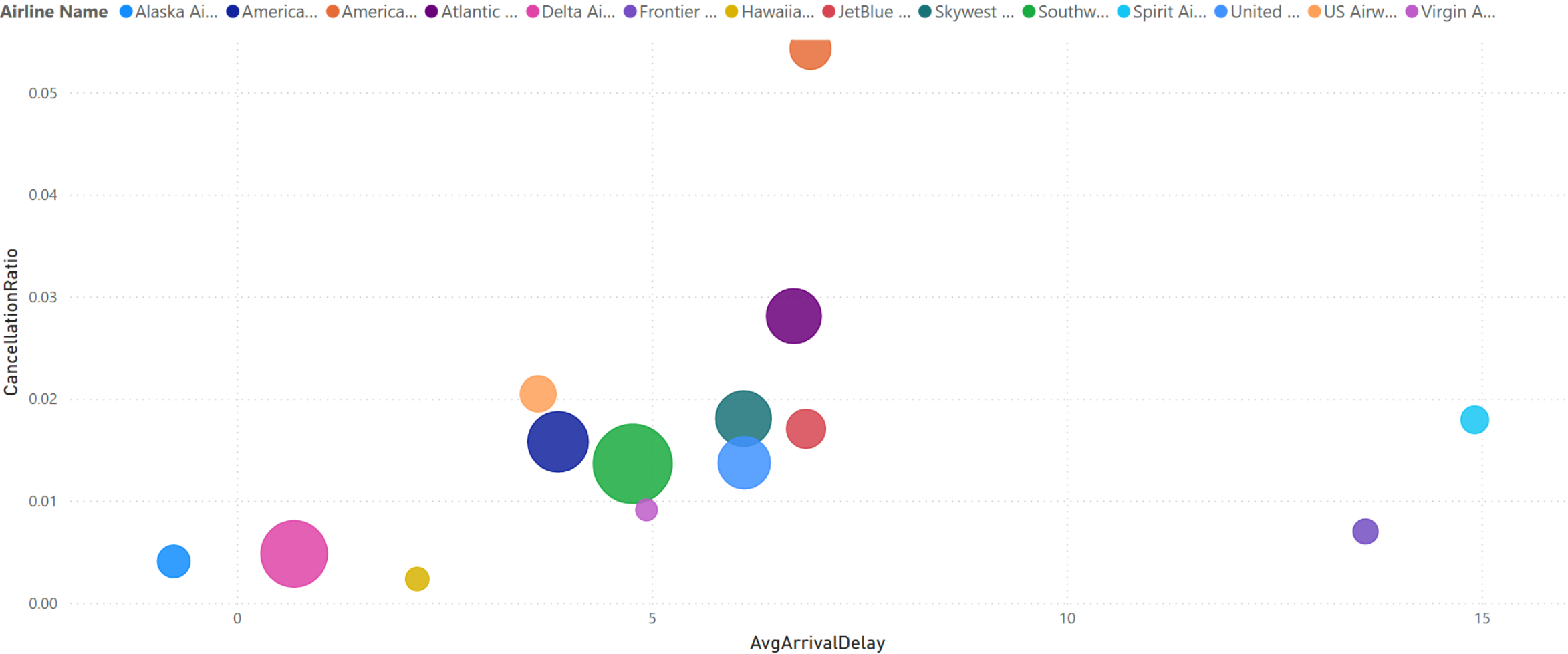


Źródło danych meteorologicznych: Iowa Environmental Mesonet (IEM)

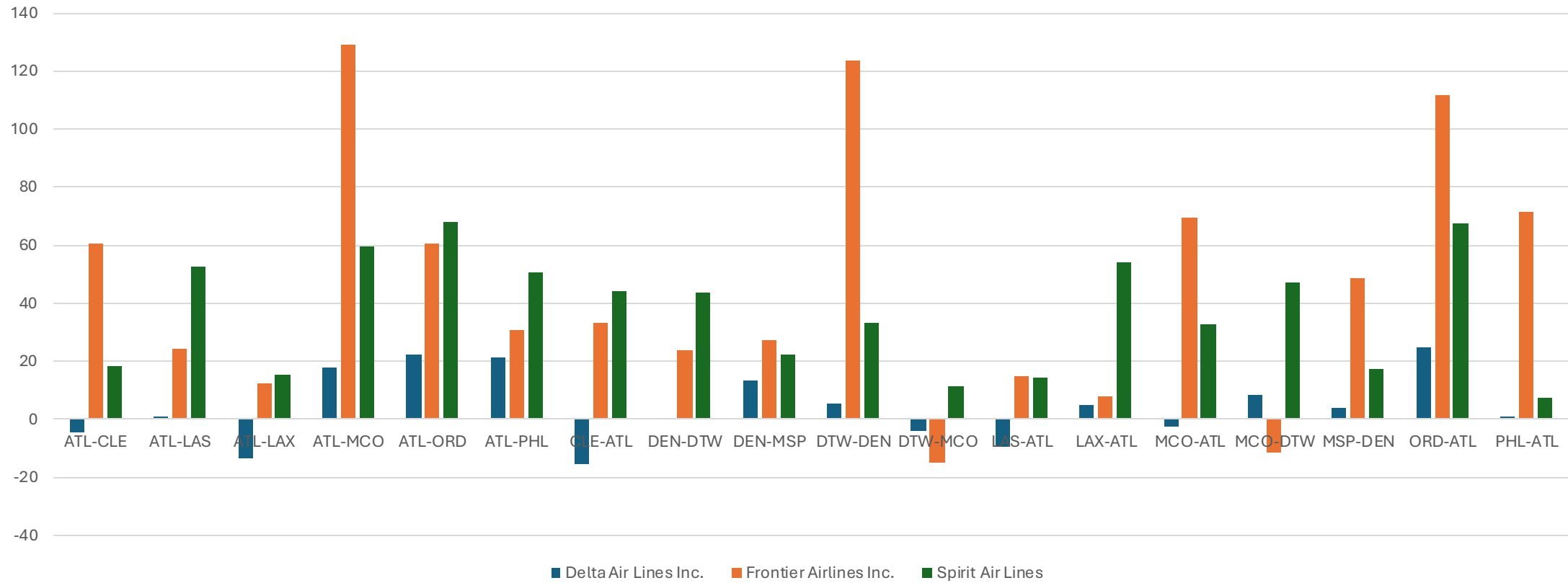


AvgArrivalDelay	Column Labels				
Row Labels	Fall	Spring	Summer	Winter	Grand Total
Alaska Airlines Inc.	-1.85	-1.76	0.95	-0.92	-0.76
American Airlines Inc.	-0.32	5.47	4.46	5.64	3.87
American Eagle Airlines Inc.	-0.96	6.91	4.97	13.01	6.91
Atlantic Southeast Airlines	2.55	6.25	8.26	8.18	6.71
Delta Air Lines Inc.	-3.43	0.77	2.08	1.96	0.69
Frontier Airlines Inc.	6.00	14.96	14.16	17.28	13.60
Hawaiian Airlines Inc.	0.53	1.44	3.49	2.56	2.18
JetBlue Airways	2.46	4.20	6.73	12.75	6.86
Skywest Airlines Inc.	3.18	3.47	6.76	10.22	6.11
Southwest Airlines Co.	-0.15	3.92	8.83	4.56	4.77
Spirit Air Lines	7.76	13.64	22.47	13.15	14.92
United Air Lines Inc.	-0.74	5.32	10.53	6.76	6.11
US Airways Inc.		2.19	5.74	4.79	3.63
Virgin America	0.09	4.49	5.97	7.80	4.94
Grand Total	0.16	4.13	6.74	6.52	4.80

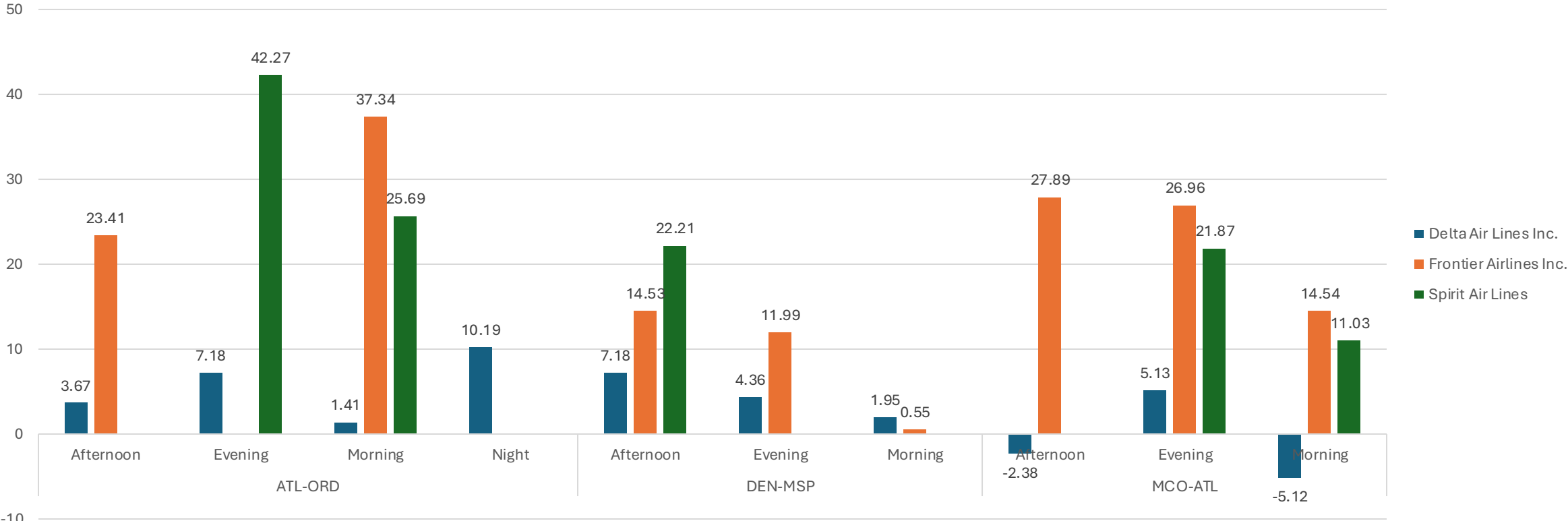
AvgArrivalDelay, CancellationRatio and FlightCount by Airline Name



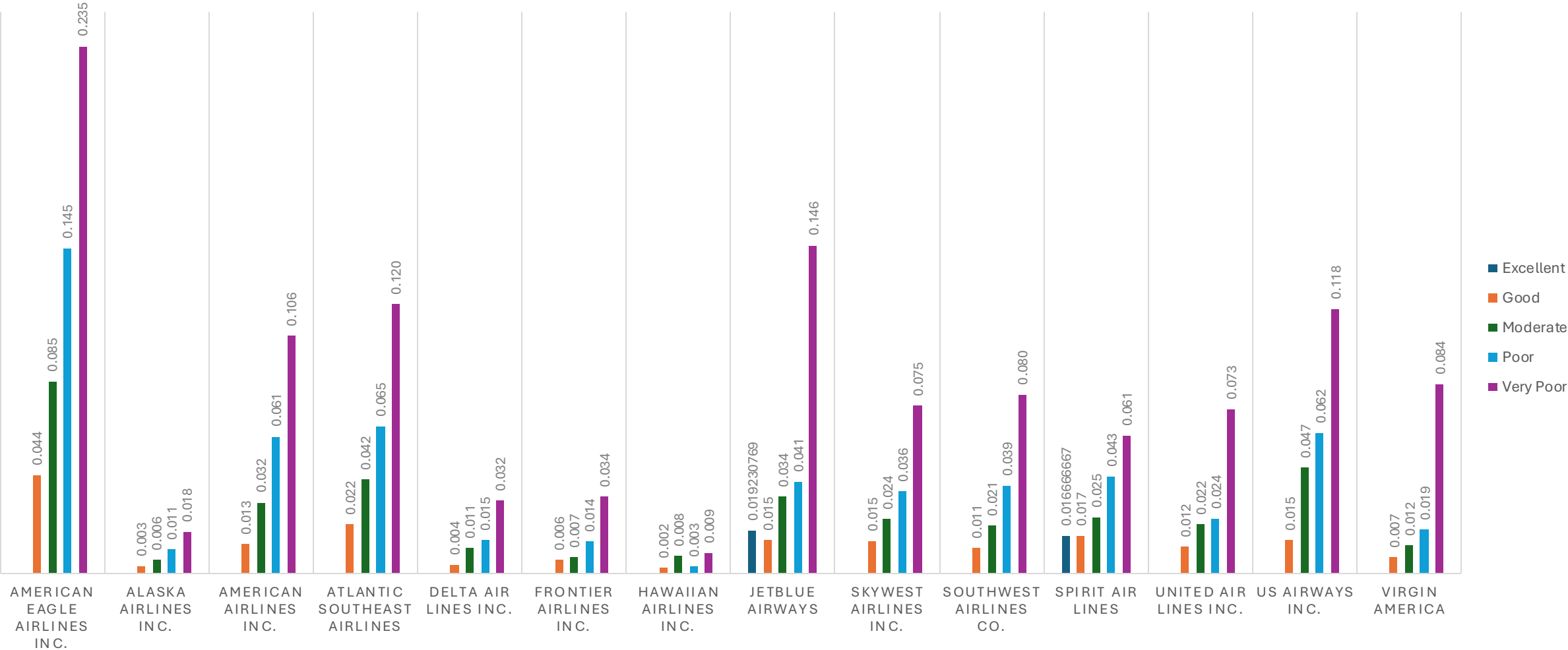
AvgArrivalDelay on common routes



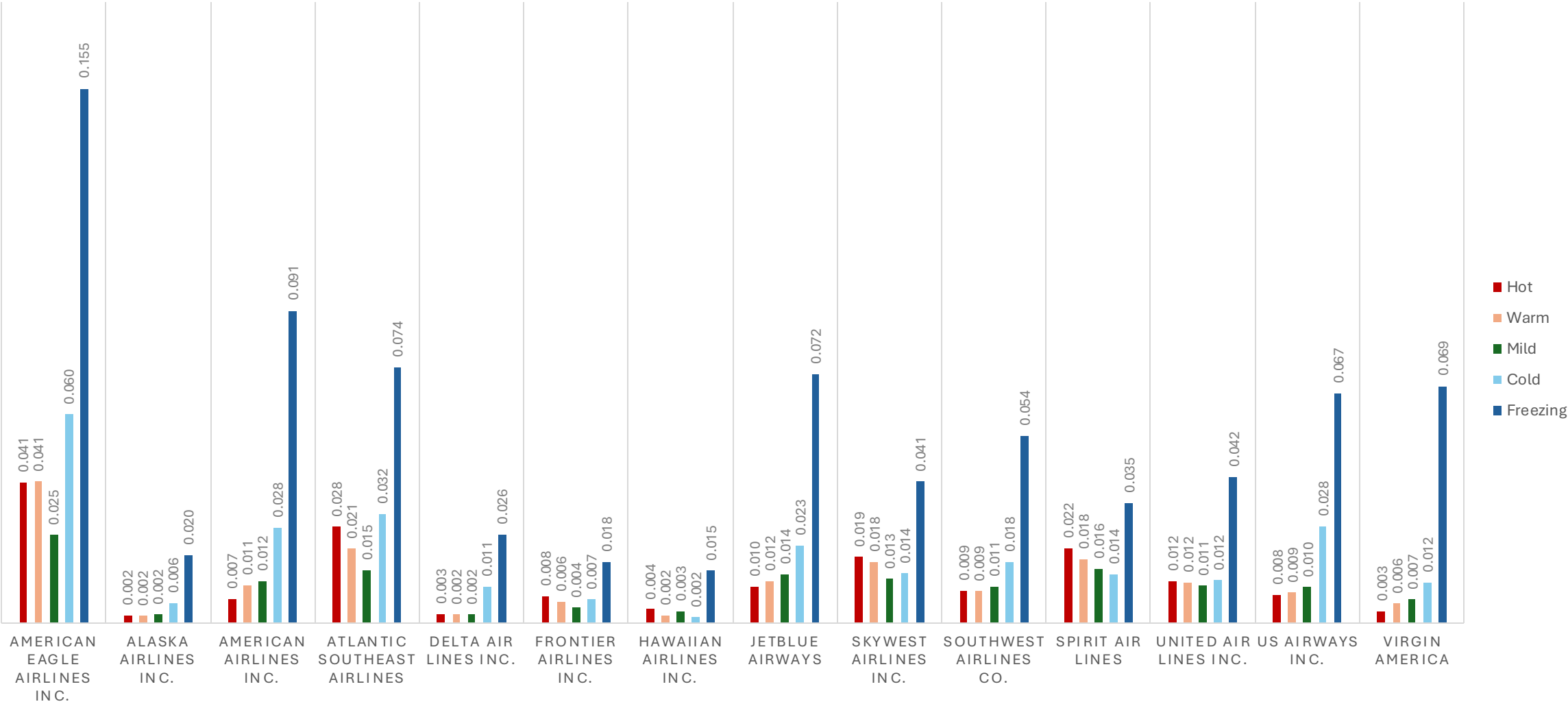
AvgArrivalDelay on Common Routes at Different Times of Day



VISIBILITY AND CANCELLATION RATIO FOR DIFFRENT AIRLINES



TEMPERATURE AND CANCELLATION RATIO FOR DIFFRENT AIRLINES



OnTimePerformanceRatio	Column Labels 					
Row Labels 	Clear	Fog	Rain	Snow	Thunderstorm	Grand Total
Alaska Airlines Inc.	0.88	0.85	0.84	0.78	0.74	0.88
American Airlines Inc.	0.83	0.78	0.73	0.64	0.68	0.82
American Eagle Airlines Inc.	0.80	0.70	0.71	0.56	0.73	0.78
Atlantic Southeast Airlines	0.82	0.77	0.73	0.64	0.65	0.81
Delta Air Lines Inc.	0.87	0.84	0.80	0.64	0.58	0.86
Frontier Airlines Inc.	0.74	0.69	0.65	0.40	0.65	0.73
Hawaiian Airlines Inc.	0.89	0.72	0.84	0.20	0.65	0.88
JetBlue Airways	0.79	0.78	0.68	0.58	0.61	0.78
Skywest Airlines Inc.	0.83	0.76	0.77	0.56	0.72	0.82
Southwest Airlines Co.	0.82	0.80	0.75	0.70	0.63	0.81
Spirit Air Lines	0.71	0.65	0.62	0.47	0.61	0.70
United Air Lines Inc.	0.80	0.76	0.70	0.58	0.61	0.79
US Airways Inc.	0.83	0.79	0.77	0.64	0.64	0.82
Virgin America	0.83	0.78	0.73	0.73	0.67	0.82
Grand Total	0.83	0.78	0.75	0.62	0.64	0.82

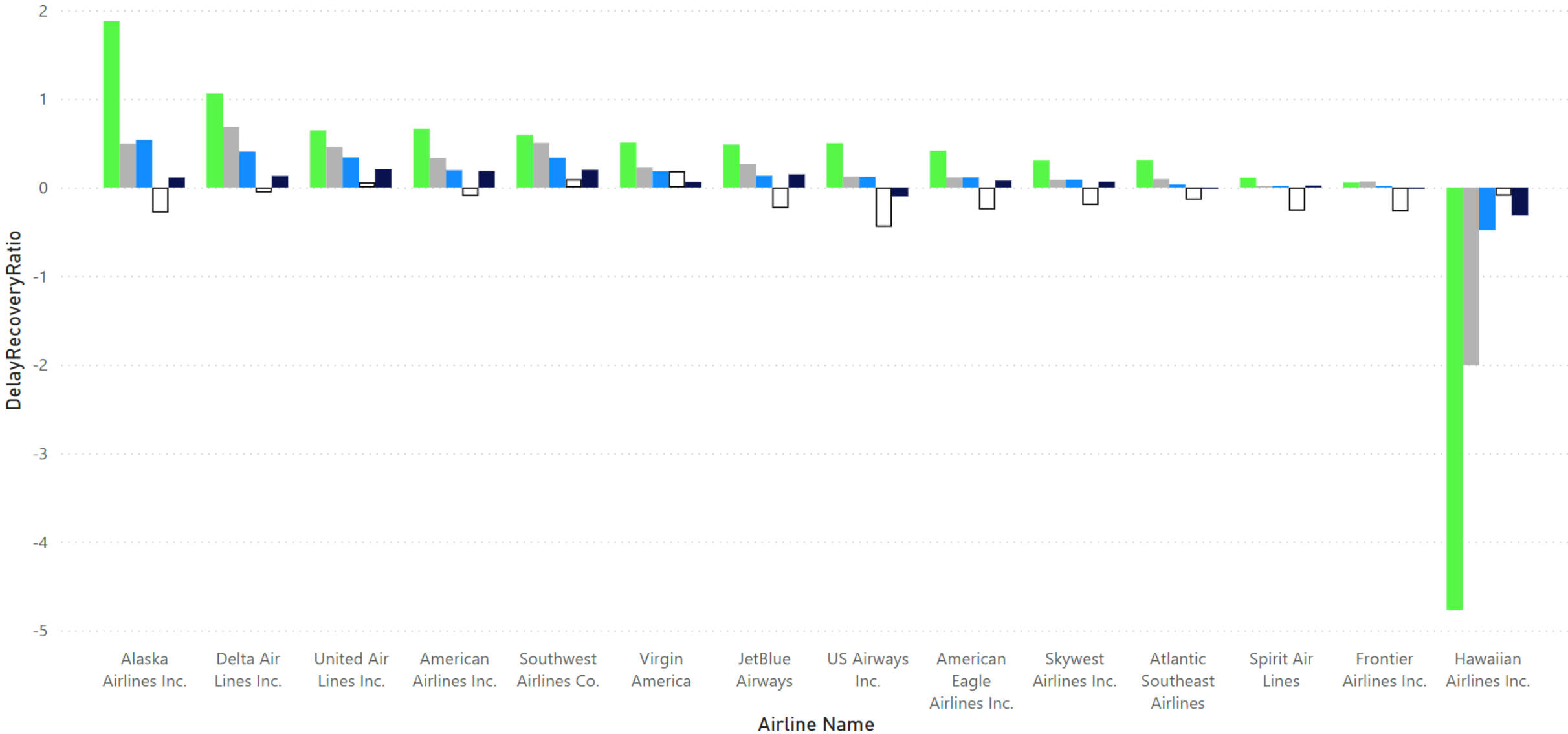
DelayRecoveryRatio

To wskaźnik, który mówi, ile z opóźnienia na starcie udało się nadrobić do przylotu

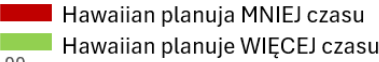
Wartość wskaźnika	Co oznacza
> 1.0	Nadrobiono wszystko i przyleciał przed czasem
1.0	Całe opóźnienie nadrobione
0.0	Nic nie nadrobiono (tyle samo opóźnienia przy lądowaniu co przy starcie)
< 0.0	Opóźnienie się pogorszyło

DelayRecoveryRatio by Airline Name and Weather Group

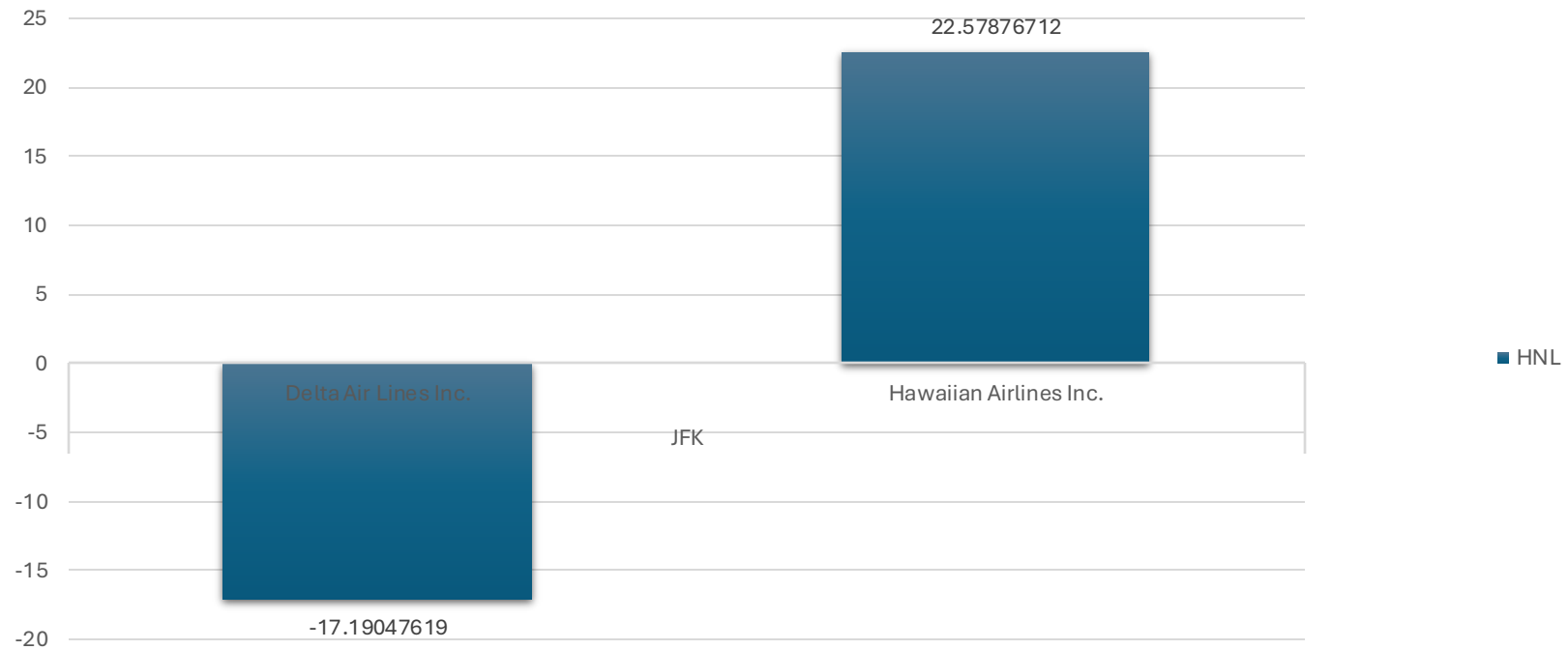
Weather Group ● Clear ● Fog ● Rain ○ Snow ● Thunderstorm



HAWAIIAN AIRLINES VS DELTA AIR LINES



Avg Arrival Delay on JFK -> HNL for Delta Air Lines and Hawaiian Airlines



Row Labels	AvgDepartureDelay	AvgArrivalDelay	DelayRecoveryRatio
Alaska Airlines Inc.	1.91	-0.54	1.28
American Airlines Inc.	9.64	4.29	0.56
American Eagle Airlines Inc.	10.72	8.07	0.25
Atlantic Southeast Airlines	8.88	6.96	0.22
Delta Air Lines Inc.	7.80	0.92	0.88
Frontier Airlines Inc.	14.22	14.40	-0.01
Hawaiian Airlines Inc.	0.85	3.09	-2.62
JetBlue Airways	11.40	6.62	0.42
Skywest Airlines Inc.	7.72	5.75	0.26
Southwest Airlines Co.	10.98	4.97	0.55
Spirit Air Lines	17.60	16.10	0.09
United Air Lines Inc.	15.06	6.67	0.56
US Airways Inc.	6.03	3.63	0.40
Virgin America	8.44	4.48	0.47
Grand Total	9.70	5.06	0.48

Number of Flights based on Temperature Group

